

Submission Worksheet

Submission Data

Course: IT202-007-F2025

Assignment: IT202 Module 4 SQL Challenge

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Status: Submitted | **Worksheet Progress:** 100+%

Potential Grade: 11.00/10.00 (110.00%)

Received Grade: 0.00/10.00 (0.00%)

Started: 10/11/2025 12:05:00 PM

Updated: 10/13/2025 3:58:33 PM

Grading Link: <https://learn.ethereallab.app/assignment/v3/IT202-007-F2025/it202-module-4-sql-challenge/grading/rt524>

View Link: <https://learn.ethereallab.app/assignment/v3/IT202-007-F2025/it202-module-4-sql-challenge/view/rt524>

Instructions

- Overview Link: <https://youtu.be/oW7AK39mgmo>
- 1. Ensure you read all instructions and objectives before starting.
- 2. Create a new branch from dev called M4-Homework
 - 1. `git checkout dev` (ensure proper starting branch)
 - 2. `git pull origin dev` (ensure history is up to date)
 - 3. `git checkout -b M4-Homework` (create and switch to branch)
- 3. Copy the template code from here: [GitHub Repository - M4 Homework](#)
 - Put all into an M4 folder inside your `public_html`
 - Expected files
 - `sql/000_create_table_todos.sql`
 - `sql/init_db.php`
 - `todos/create.php`
 - `todos/pending.php`
 - `todos/completed.php`
 - `nav.php`
 - Immediately record to history
 - `git add public_html`
 - `git commit -m "adding M4 HW baseline files"`
 - `git push origin M4-Homework`
 - Create a Pull Request from M4-Homework to dev and keep it open
- 4. Fill out the below worksheet
 - Each Problem requires the following as you work
 - Ensure there's a comment with your UCID, date, and brief summary of how the problem was solved
 - Update `ucid` in `nav.php` where noted
 - Code solution (add/commit periodically as needed) (refer to comments in the create, pending, completed files)

Field names are task, due, and assigned. These items were named to match the columns in the SQL table. "Self" was default for assigned input by using value="self" so that the text defaults to this value.

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≡ Task #2 (0.75 pts.) - Form Submission PHP Validations

Progress: 100%

Part 1:

Progress: 100%

Details:

- Validate the incoming data for correct format based on the SQL table definition.
- When not valid, provide a user-friendly message of what specifically was wrong and set `$is_valid` to false
- Assigned should check for "self" if a valid format/value isn't provided; it should default to "self" if all checks for this property fail
- Include screenshot(s) from Heroku Dev demonstrating each validation (ensure URL is visible)

```
1 <?php if (empty($_GET)) {  
    // done variables, don't edit  
    $task = $_GET["task"];  
    $due = $_GET["due"]; //must be a valid mysql date format  
    $assigned = $_GET["assigned"]; // Must be "self" or a valid format (not empty or equivalent)  
  
    $is_valid = true;  
    // YOU validate the incoming data for correct format based on the SQL table definition.  
    // When not valid, provide a user-friendly message of what specifically was wrong and set $is_valid to false  
    // Assigned should check for "self" if a valid format/value isn't provided.  
    // Start validation here.  
    // can edit here  
  
    // echo 10/11/25  
    // check if due is a string long (likely in DATE format) and if assigned is empty/task  
    if (strlen($due) != 10) {  
        echo "Date is not valid. Please provide a valid date then";  
        $is_valid = false;  
    }  
    if (empty(trim($assigned))) {  
        echo "Assigned is not valid. Defaulting to 'self'";  
        $assigned = "self";  
        $is_valid = false;  
    }  
    // End validations  
    if ($is_valid) {
```

php validation

rt524-it202-007-dev-5d0617ee62f2.herokuapp.com/M4/todos/create.php?task=taskname&due=10101-01-01&assigned=+++++

Date is not valid. Please provide a valid date.
Assigned is not valid. Defaulting to 'self'

Create ToDo Pending ToDo Completed ToDo

rt524 | Monday, October 13, 2025 at 3:31 PM

Create ToDo

Task:

Due:

Assigned:

show validation errors

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≡ Part 2:

Progress: 100%

Details:

Briefly explain your validation logic.

Your Response:

if the date is not the valid date length (10 characters), it will throw a message saying that the date is not valid. The HTML calendar will provide a valid SQL date but has space for a few extra characters which would not work, so the limit would always be 10 characters. For assigned, if it is made of up entirely spaces, it will default to "self"



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≡ Task #3 (0.75 pts.) - Show the code snippet related to the INSERT query

Progress: 100%

Part 1:

Progress: 100%

Details:

Design a query to insert the incoming data to the proper columns. Ensure valid and proper PDO named placeholders are used. <https://phpdelusions.net/pdo>

```

<?php ($is_valid) {
    //
    Design a query to insert the incoming data to the proper columns.
    Ensure valid and proper PDO named placeholders are used.
    https://phpdelusions.net/pdo

    //SQL 10/11/25 use INSERT INTO tablename (col1, col2, col3) VALUES (task, due, assigned) for query
    // In param, define the array for the array
    //
    $query = "INSERT INTO M4 Table (task, due, assigned) VALUES (:task, :due, :assigned)"; // col1: 01,
    $params = [
        ':task' => $task,
        ':due' => $due,
        ':assigned' => $assigned
    ]; // Apply the proper PDO placeholder to variable mapping here
    try {
        $db = getDB();
        $stmt = $db->prepare($query);
        $r = $db->execute($stmt);
        if ($r) {
            echo "Inserted new Task with id " . $db->lastInsertId();
        } else {
            echo "Failed to insert";
        }
    } catch (PDOException $e) {
        // echo "Error";
        // Check if the exception was related to a unique constraint
        // provide an appropriate user friendly message for this scenario
        // otherwise show the default message below
    }
}

```

SQL insert query



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Part 2:

Progress: 100%

Details:

Briefly describe **how** the query works in plain words.

Your Response:

Inserts the task, due date, and assigned into their respective columns in the table. The parameters

are created in an associative array with their corresponding values and executed after preparing the query.

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Task #4 (+ 1.01 pts.) - Extra Credit: Handle Unique Constraint Errors

Progress: 100%

Part 1:

Progress: 100%

Details:

Show the code related to handling the unique constraint errors and the user-friendly message

```
} catch (PDOException $e) {  
    // extra credit  
    // check if the exception was related to a unique constraint  
    // provide an appropriate user-friendly message for this scenario  
    // Otherwise show the default message below  
  
    // r1524 10/11/25 find the duplicate primary key error message. if it is duplicate, show error message  
    // otherwise, throw default error  
    if ($e->getCode() == 23000) {  
        echo "Duplicate entry. Ensure that for each task of the same name, the due date is unique.";  
    }  
    else {  
        echo "There was an error inserting the record; check the logs (terminal)";  
        echo $e->getMessage();  
        error_log("Insert Error: " . var_export($e, true)); // shows in the terminal  
    }  
}  
} else {  
    error_log("Creation input wasn't valid");  
}
```

unique constraint error code

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Part 2:

Progress: 100%

Details:

Briefly explain **how** your code solves the challenge.

Your Response:

the code looks for if there are any errors when inserting into the table. if an error is found, it spits out the proper message to the user, in this case a duplicate value which the table would not accept. otherwise, gives a default

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Task #5 (0.75 pts.) - Create Page URLs

Progress: 100%

Details:

- Direct link to the file in the homework-related branch from Github (should end in `.php`)
- Direct link to the file on Heroku Prod (Just grab the base prod url and manually enter the path to the file)

[illegible]


```

SELECT * FROM TABLE1
WHERE ID = 1 AND IL COMPLETED = 0
UNION ALL
SELECT * FROM TABLE2
WHERE ID = 1 AND IL COMPLETED = 0

```



```
...
    $task['completed'] = true;
    $task['timestamp'] = time();
    $db->update($table, $task, $id);
    echo "Task #$id completed!\n";
}

// Update the task with the ID provided
function updateTask($id) {
    $task = $db->getTask($id);
    if ($task) {
        $task['completed'] = true;
        $task['timestamp'] = time();
        $db->update($table, $task, $id);
        echo "Task #$id completed!\n";
    } else {
        echo "Task #$id not found!\n";
    }
}

// Main function to handle the request
function main() {
    $id = $_GET['id'];
    if ($id) {
        updateTask($id);
    } else {
        echo "No ID provided!\n";
    }
}

main();
```

query for updating todos

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⇒ Part 2:

Progress: 100%

Details:

Briefly describe the query works in plain words.

Your Response:

the query updates the existing table by marking the specific task with the ID was complete and adding the timestamp of when it was complete to the table

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⇒ Task #3 (1 pt.) - Pending Page URLs

Progress: 100%

Details:

- Direct link to the file in the homework-related branch from GitHub (should end in `.php`)
- Direct link to the file on Heroku Prod (Just grab the base prod url and manually enter the path to the file)

URL #1

https://github.com/rt-524/rt524-it202-007/M4-Homework/public_html/M4/todos/pending.php



<https://github.com/rt-524/rt5>



URL #2

<https://rt524-it202-007-prod-a195f178bfec.herokuapp.com/M4/todos/pending.php>



<https://rt524-it202-007-prod-i>



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Task #1 (1 pt.) - Fetching Todos

Progress: 100%

Part 1:

Progress: 100%

Details:

- Refer to the HTML table below and build a query that'll select the columns in the same order as the table from the Todo table
- Cross-reference the HTML table columns with what'd most plausibly match the SQL table, aside from the notes below
- For the completed date, you'll need to extract the date portion from the completed column
- For the Status part, you'll need to calculate the "days_offset" from the completed date, ensure the virtual column matches "days_offset"
- Filter the results where the todo item is completed and order the results by most recently completed and most recently due
- No limit is required
- Include screenshot(s) from Heroku Dev demonstrating 5 different records (ensure URL is visible)

```

/* Refer to the HTML table below and build a query that'll select the columns in the same order as the table from the Todo table.
Cross-reference the HTML table columns with what'd most plausibly match the SQL table aside from the notes below.
For the completed date you'll need to extract the date portion from the completed column.
For the Status part, you'll need to calculate the "days_offset" from the completed date, ensure the virtual column matches "days_offset".
Filter the results where the todo item is completed and order the results by most recently completed and most recently due.
No limit is required.
ETODO 10/13/25 same query as pending but where is_complete is 1 and sort by the completed timestamp
*/
$query = "SELECT id, task, due, completed, DATEDIFF(due, CURRENT_TIMESTAMP) AS days_offset, assigned
FROM my_todo
WHERE is_complete = 1
ORDER BY completed DESC"; // edit this
$results = [];
try {
    $stmt = $db->prepare($query);
    $r = $stmt->execute();
    if ($r) {
        $results = $stmt->fetchAll();
    }
} catch (PDOException $e) {
    echo "Error fetching pending todos: check the logs. (terminal)";
    error_log("Select Error: " . var_export($e, true)); // shows in the terminal
}
}

```

sql query of complete

Completed Todos

ID	Task	Due Date	Completed Date	Status	Assigned
10	task4	2025-10-29	2025-10-13 19:30:11	Completed (16 days)	oBao
10	task1	2025-10-15	2025-10-13 19:30:10	Completed (2 days)	oBf
11	task3	2025-10-18	2025-10-13 19:30:10	Completed (5 days)	oBf
11	task2	2025-10-13	2025-10-13 19:30:09	Completed (0 days)	oBao
12	task5	2025-10-27	2025-10-13 19:30:08	Overdue by 14 days	oBf
9	task	2025-10-15	2025-10-13 19:30:07	Completed (2 days)	oBf
1	task	2025-10-23	2025-10-13 08:00:00	Completed (10 days)	oBf
4	task	2025-10-11	2025-10-13 08:00:00	Overdue by 2 days	oBf
7	task	2025-10-12	2025-10-13 08:00:00	Overdue by 1 day	oBf
8	task	2025-10-14	2025-10-13 08:00:00	Completed (1 day)	oBao

heroku dev complete records



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Part 2:

Progress: 100%

Details:

Briefly describe **how** the query works in plain words.

Your Response:

the query selects the required columns from the table and finds items that have been marked as complete, displaying them in descending order from when they were marked as complete.



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Task #2 (1 pt.) - Completed Page URLs

Progress: 100%

Details:

- Direct link to the file in the homework-related branch from GitHub (should end in **.php**)
- Direct link to the file on Heroku Prod (Just grab the base prod url and manually enter the path to the file)

URL #1

https://github.com/rt-524/rt524-it202-007/M4-Homework/public_html/M4/todos/completed.php



URL

https://github.com/rt-524/rt524-it202-007/M4-Homework/public_html/M4/todos/completed.php

**URL #2**

<https://rt524-it202-007-prod-a195f178bfec.herokuapp.com/M4/todos/completed.php>



URL

<https://rt524-it202-007-prod-a195f178bfec.herokuapp.com/M4/todos/completed.php>



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Section #5: (1 pt.) Misc

Progress: 100%

Task #1 (0.33 pts.) - Github Details

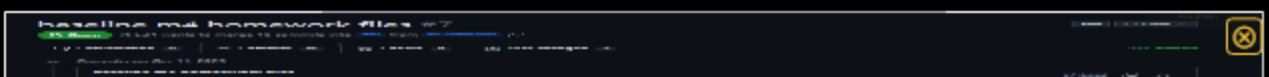
Progress: 100%

Part 1:

Progress: 100%

Details:

From the Commits tab of the Pull Request screenshot the commit history



commit 177	177	177
commit 176	176	176
commit 175	175	175
commit 174	174	174
commit 173	173	173
commit 172	172	172
commit 171	171	171
commit 170	170	170
commit 169	169	169
commit 168	168	168
commit 167	167	167
commit 166	166	166
commit 165	165	165
commit 164	164	164
commit 163	163	163
commit 162	162	162
commit 161	161	161
commit 160	160	160
commit 159	159	159
commit 158	158	158
commit 157	157	157
commit 156	156	156
commit 155	155	155
commit 154	154	154
commit 153	153	153
commit 152	152	152
commit 151	151	151
commit 150	150	150
commit 149	149	149
commit 148	148	148
commit 147	147	147
commit 146	146	146
commit 145	145	145
commit 144	144	144
commit 143	143	143
commit 142	142	142
commit 141	141	141
commit 140	140	140
commit 139	139	139
commit 138	138	138
commit 137	137	137
commit 136	136	136
commit 135	135	135
commit 134	134	134
commit 133	133	133
commit 132	132	132
commit 131	131	131
commit 130	130	130
commit 129	129	129
commit 128	128	128
commit 127	127	127
commit 126	126	126
commit 125	125	125
commit 124	124	124
commit 123	123	123
commit 122	122	122
commit 121	121	121
commit 120	120	120
commit 119	119	119
commit 118	118	118
commit 117	117	117
commit 116	116	116
commit 115	115	115
commit 114	114	114
commit 113	113	113
commit 112	112	112
commit 111	111	111
commit 110	110	110
commit 109	109	109
commit 108	108	108
commit 107	107	107
commit 106	106	106
commit 105	105	105
commit 104	104	104
commit 103	103	103
commit 102	102	102
commit 101	101	101
commit 100	100	100

commit history



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Part 2:

Progress: 100%

Details:

Include the link to the Pull Request (should end in `/pull/#`)

URL #1

<https://github.com/rt-524/rt524-it202-007/commits>



URL

<https://github.com/rt-524/rt524-it202-007/pull/1>



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Task #2 (0.33 pts.) - WakaTime - Activity

Progress: 100%

Details:

- Visit the WakaTime.com Dashboard
- Click `Projects` and find your repository
- Capture the overall time at the top that includes the repository name
- Capture the individual time at the bottom that includes the file time
- Note: The duration isn't relevant for the grade and the visual graphs aren't necessary

[Projects](#) • rt524-it202-007

3 hrs 6 mins over the [Last 7 Days](#) in rt524-it202-007 under [all](#) branches. 📊

waketime overall

Files		Branches	
1 hr 27 mins	public_html/M4/todos/create.php	3 hrs 6 mins	M4-Homework
44 mins	public_html/M4/todos/pending.php		
30 mins	public_html/M4/todos/completed.php		
7 mins	public_html/M4/sql/init_db.php		
5 mins	public_html/M4/nav.php		
5 mins	..ers/rosha/Downloads/Quantifiers.java		
3 mins	public_html/test_db.php		
1 min	public_html/M2/problem1.php		
20 secs	public_html/index.php		
18 secs	..html/M4/sql/000_create_table_todo.sql		
2 secs	public_html/README.md		

wakatime files and branches



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☰ Task #3 (0.33 pts.) - Reflection

Progress: 100%

⇒ Task #1 (0.33 pts.) - What did you learn?

Progress: 100%

Details:

Briefly answer the question (at least a few decent sentences)

Your Response:

I learned how PHP is able to work hand in hand with SQL as well as how to construct queries. I did not think that integrations with SQL would be as simple as this and easy to do through vscode as I thought databases were much more complex than they really are.



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⇒ Task #2 (0.33 pts.) - What was the easiest part of the assignment?

Progress: 100%

Details:

Briefly answer the question (at least a few decent sentences)

Your Response:

The easiest part of the assignment was building the queries. I have had a little exposure to SQL before and I am currently taking a database class which has helped me for those parts.



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⇒ Task #3 (0.33 pts.) - What was the hardest part of the assignment?

Progress: 100%

Details:

Briefly answer the question (at least a few decent sentences)

Your Response:

The hardest parts of the assignment were figuring out the available methods from SQL to deal with dates as well as understanding how PDO worked. Some parts like the associative array were confusing at first, especially with the use of the colon.



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